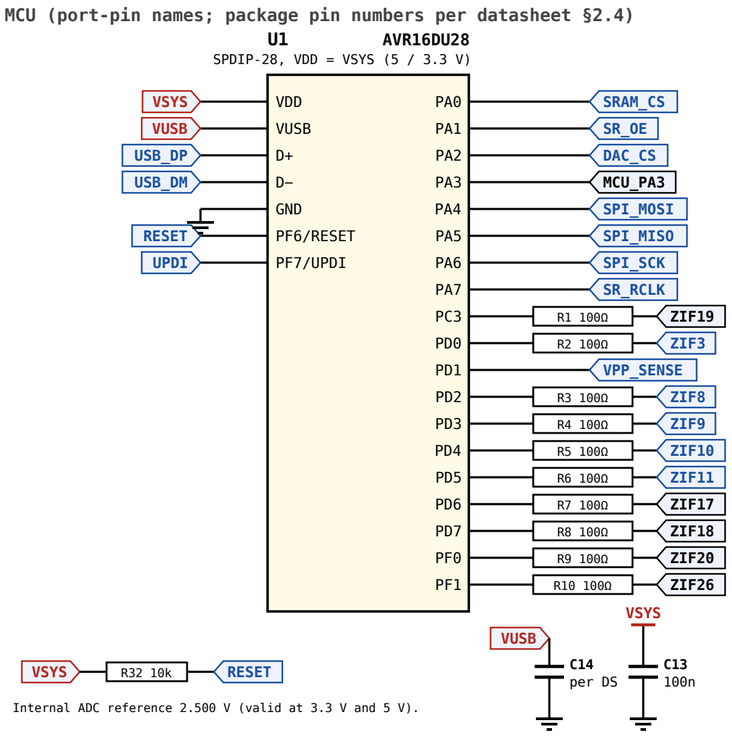
J48 GAL / PAL Programmer — Schematic (Rev I: J5 expansion header)

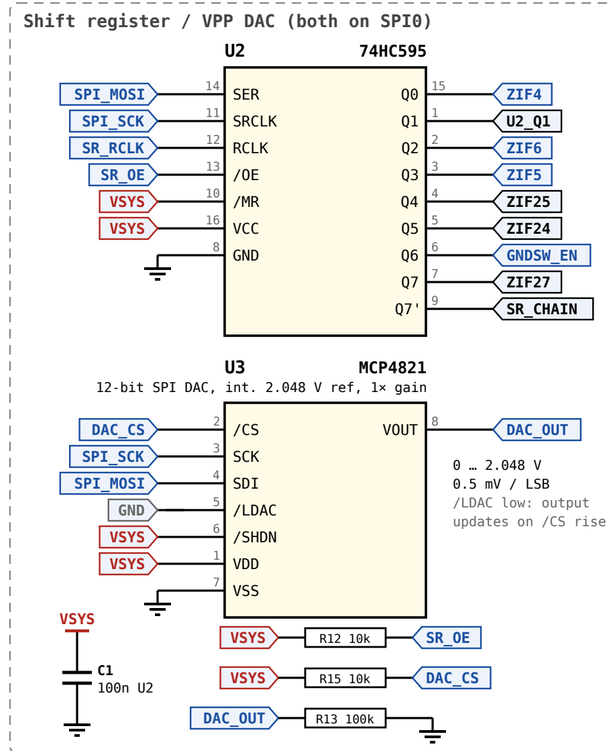
Native USB · SPI: 2× 74HC595, MCP4821 DAC, 23LC1024 SRAM · VSYS = 5 V / 3.3 V (J4) · ZIF-28 · J5 expansion · purple = changed in Rev I.

MCU (port-pin names; package pin numbers per datasheet §2.4)

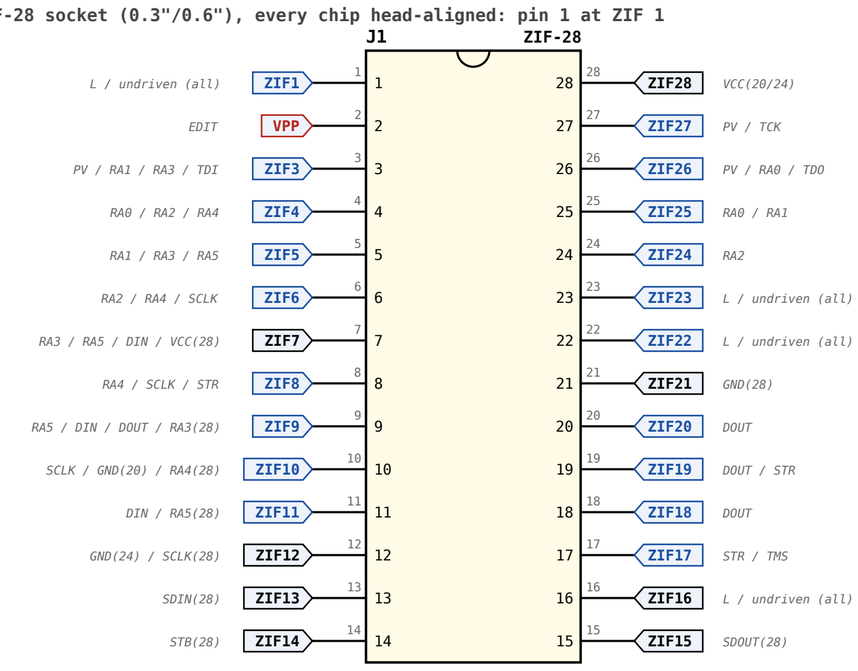


Internal ADC reference 2.500 V (valid at 3.3 V and 5 V).

Shift register / VPP DAC (both on SPI0)



ZIF-28 socket (0.3"/0.6"), every chip head-aligned: pin 1 at ZIF 1

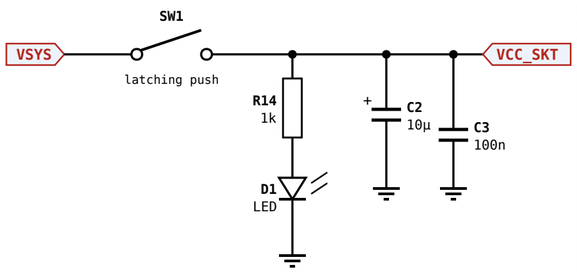


grey text = programming role(s); (20)/(24)/(28) = only that chip size

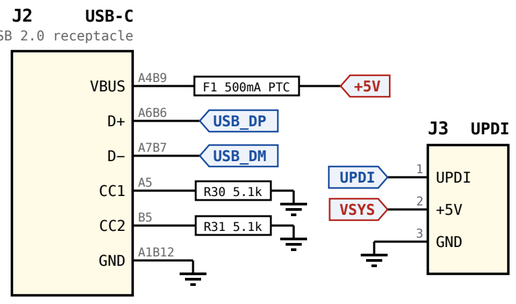
20-pin: ZIF 1–10, 19–28 · 24-pin: ZIF 1–12, 17–28 · 28-pin: ZIF 1–28

GND/VCC are switched per size (see ZIF-28 switching, bottom)

USB, socket power & UPDI

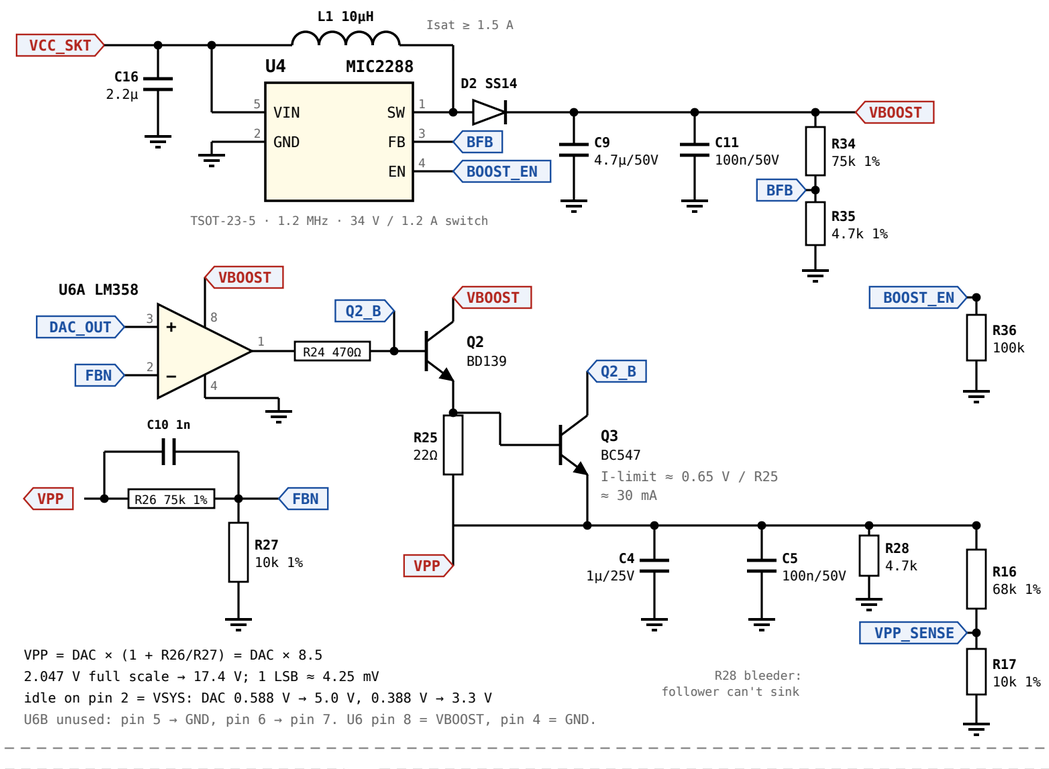


SW1: VSYS → VCC_SKT. OFF to insert/remove. C2/C3 by Q6/Q7.
SW1 also removes VBOOST, so VPP collapses to 0 V.



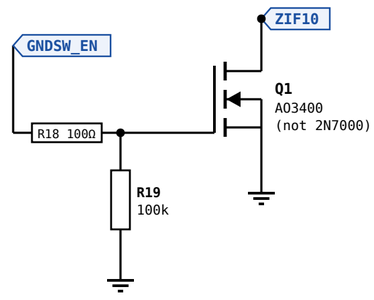
C12 10µ bulk on +5V near J2

VPP generator: MIC2288 boost → DAC → op-amp → pass transistor (0 … 17.4 V)



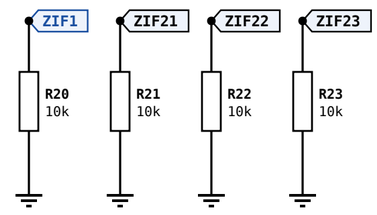
VPP = DAC × (1 + R26/R27) = DAC × 8.5
2.047 V full scale → 17.4 V; 1 LSB ≈ 4.25 mV
idle on pin 2 = VSYS: DAC 0.588 V → 5.0 V, 0.388 V → 3.3 V
U6B unused: pin 5 → GND, pin 6 → pin 7. U6 pin 8 = VBOOST, pin 4 = GND.

ZIF-10 ground switch



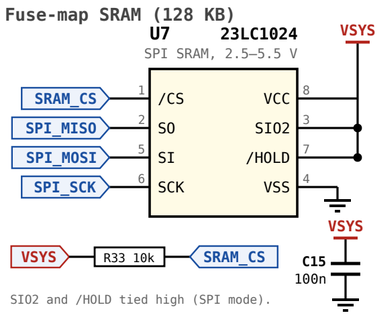
Driven by U2 Q6; ON only for 20-pin chips.
At reset U2 outputs are off (/OE pulled high by R12) and R19 holds Q1 off.

Pull-downs (ZIF 1, 21–23)



Low / undriven for every device.
Resistors, not hard ground, so a chip driving one can't short.

Fuse-map SRAM (128 KB)



SIO2 and /HOLD tied high (SPI mode).

SPI0 bus (mode 0): who is selected by what

Device	Select line	Data	Notes
U2+U9 74HC595	PA7 latch (RCLK)	MOSI, SCK	16-bit chain, U9 byte first
U3 MCP4821	PA2 /CS (R15 up)	MOSI, SCK	write-only, 16-bit frame; 20 MHz max
U7 23LC1024	PA0 /CS (R33 up)	MOSI, MISO, SCK	fuse map; sequential mode default

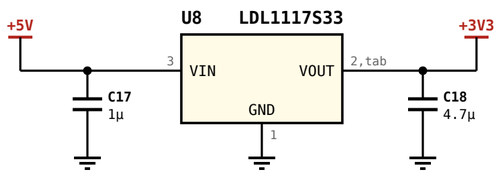
The 595 chain sees every SPI byte, but its outputs move only when PA7 is pulsed, so firmware sends both 595 bytes (U9 first) right before each latch.
R13 pulls DAC_OUT to 0 V whenever U3 is in shutdown or unwritten, so VPP starts at 0 V; firmware writes the idle code (1176 @5 V, 776 @3.3 V) at boot.
Max SCK 12 MHz at 5 V; at 3.3 V keep SCK ≤ 8 MHz for the 74HC595.

Notes

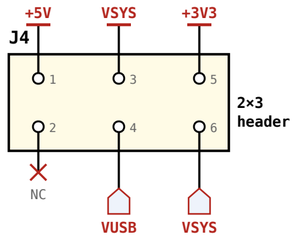
1. Net labels with the same name are connected. ZIF pin roles come from the device pinout spreadsheet.
2. Port-pin names only: take SPDIP/SSOP pin numbers from the AVR DU datasheet (DS40002576, §2.4).
3. SPI0: PA4 MOSI, PA5 MISO, PA6 SCK; PA7 = 595 latch (GPIO; set SPI0 CTRLB.SSD). No I2C in this revision.
4. Each SPI device has its own select (table, bottom left); /CS pull-ups keep U3 and U7 deselected at reset.
5. No crystal: PA0/PA1 and PF0/PF1 are used as GPIO. Confirm crystal-less USB in the datasheet.
6. C14 (VUSB) value and any D+/D− series resistors: per the datasheet's USB section.
7. MCP4821 internal reference: trim the DAC code against the PD1 ADC readback (internal 2.500 V ref).
8. Check the LM358 loop for oscillation at 5 V and 16.5 V; adjust C10 (470p–4.7n) if needed.
9. U4: VBOOST = 1.24 V × (1 + R34/R35) ≈ 21.0 V. TSOT-23 part has no OVP: 1% resistors, inspect FB before power-up.
10. BOOST_EN (U9 Q3) low or Hi-Z (R36) = boost off: VBOOST ≈ VCC_SKT − 0.4 V, so VPP stays ≤ ~3 V.
11. U7 holds the fuse map (lifts the 2 KB limit). All 19 MCU pins are used; PA3 now reads SDOUT (ZIF15).
12. Schematic-level reference, not a verified board. Breadboard (SPDIP-28) before ordering PCBs.

J48 GAL / PAL Programmer — AVR16DU28
Rev I · Sheet 1/1 · 2026-10-18
ZIF pin map derived from Afterburner (ole00)

Logic supply select (new in Rev G): VSYS = 5 V or 3.3 V



SOT-223, 1.2 A, stable with ceramic caps.
C17/C18 values per the LDL1117 datasheet.



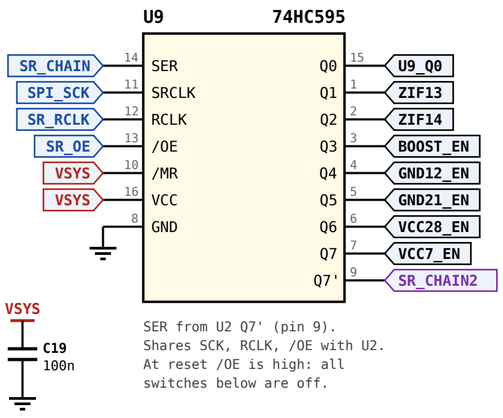
5V: shunts 1–3 + 2–4
3V3: shunts 3–5 + 4–6

Position	Shunts	VSYS	VUSB (U1 pin 6)
5V	1–3, 2–4	+5V	from U1's internal USB regulator (USBVREG = 1), as Rev F
3V3	3–5, 4–6	+3V3	tied to VDD; firmware writes SYSCFG.VUSBCTRL.USBVREG = 0

VSYS feeds: U1 VDD (pins 18, 24), U2, U3, U7, J3 pin 2, the R12/R15/R32/R33 pull-ups, and SW1 → VCC_SKT (ZIF 28 / ZIF 7 via Q6 / Q7, and the U4 boost input). +5V (VBUS after F1) feeds only J4 and U8.

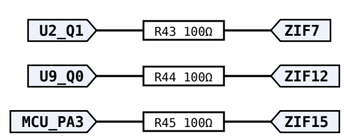
13. Move the J4 shunts only with USB unplugged, and keep both on the same side (silkscreen "5V | 3V3"). Mixed positions put the wrong voltage on VUSB.
14. 3.3 V mode is the datasheet's external-3.3 V power configuration (VDD = VUSB). USB needs VDD ≥ 3.0 V. Check C14 against that configuration.
15. No MCU pin is free for a sense line: firmware measures VDD with the ADC at boot (VDDDIV10 input; confirm the DU has it) and refuses 5 V-only parts at 3.3 V.
16. 3.3 V mode is for low-voltage parts (e.g. ATF16LV8C, ATF22LV10C) and needs Afterburner support for them. GAL16V8/20V8/22V10 and ATF16V8B/22V10C are 5 V only.
17. Firmware changes: U3 at 1× gain (GA bit = 1); VPP = DAC × 8.5; ADC on the 2.500 V reference (R16 68k keeps VPP_SENSE ≤ 2.7 V at maximum VBOOST, below VSYS).
18. Q1 must be a logic-level FET (AO3400, rated at Vgs 2.5 V). In 3.3 V mode U8 also feeds the MIC2288 (≈0.2 A at the 30 mA VPP limit): check U8's temperature.

ZIF-28 switching (new in Rev H): U9 second 595, GND and VCC switches, series resistors



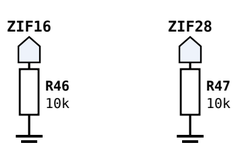
SER from U2 Q7' (pin 9).
Shares SCK, RCLK, /OE with U2.
At reset /OE is high: all switches below are off.

Series resistors

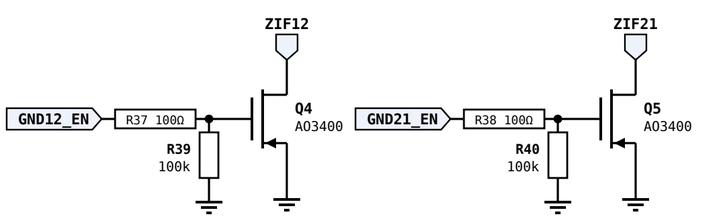


R43: U2 Q1 still drives ZIF7 (RA3 for 22V10) but can't fight Q7 when ZIF7 is the 26CV12's VCC (set Q1 high).
R44: ZIF12 is SCLK for 28-pin, GND (Q4) for 24-pin.
R45: PA3 (was BOOST_EN) reads 26CV12 SDOUT.

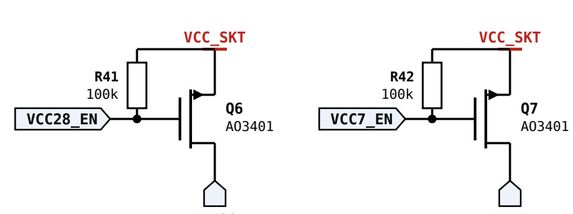
Pull-downs:



Ground switches (ON = high)



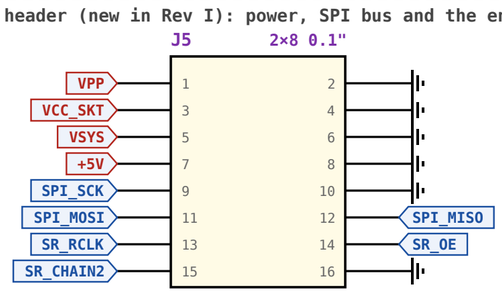
VCC switches (ON = low, P-FET)



Chip size	Q1 ZIF10	Q4 ZIF12	Q5 ZIF21	Q6 ZIF28	Q7 ZIF7	Chip GND / VCC
20-pin (16V8, 18V10)	**ON**	off	off	**ON**	off	ZIF 10 / ZIF 28
24-pin (22V10, 20V8, 6001…)	off	**ON**	off	**ON**	off	ZIF 12 / ZIF 28
28-pin (26CV12, 26V12)	off	off	**ON**	off	**ON**	ZIF 21 / ZIF 7

Firmware interlock: one GND and one VCC switch per size; all off at reset. Never Q6 with Q7; never Q4 with a 28-pin chip.

J5 expansion header (new in Rev I): power, SPI bus and the end of the 595 chain for adapter boards



Power at one end, a GND beside each rail and next to SPI_SCK.

J5 pins	Signal	Notes
1	**VPP**	Same net as ZIF pin 2: the ZIF socket must be empty when an adapter is in use.
3	**VCC_SKT**	Off when SW1 is off; the adapter switches it to its own socket.
5, 7	**VSYS, +5V**	VSYS follows J4 (5 V or 3.3 V); +5V is always 5 V for 5 V-only parts. Both stay live: plug in with USB off.
9, 11, 12	**SPI_SCK, MOSI, MISO**	Shared with U2, U3, U7 and U9. Adapter devices must release MISO when not selected.
13, 14	**SR_RCLK, SR_OE**	Latch and output enable for every 595 in the chain, on board and on the adapter.
15	**SR_CHAIN2**	End of the on-board chain (U9 Q7′). Feed the first adapter 595's SER from here.
2,4,6,8,10,16	**GND**	

Adapters: 74HC595s on the chain for outputs, 74HC589s (3-state) for inputs over MISO, an SPI EEPROM for an adapter ID. See docs/expansion.md.